Phase 1 Preprocessing Strategy Report: An Explainable Risk-Triage System for Public Procurement Oversight in Kenya

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Abstract

This report presents the data preprocessing strategy for a procurement risk-triage system built on Kenya's open contracting data. The dataset spans five linked CSV files covering 259,709 tender records, 109,163 awards, 79,870 supplier records, 109,163 contracts, and 813,078 party records published between July 2018 and August 2026. Because no ground-truth corruption labels exist, the task is formulated as unsupervised risk ranking. The target variable is a composite risk score engineered from five computable red-flag indicators. This report audits the raw data, diagnoses missingness mechanisms, and specifies justified strategies for cleaning, imputation, encoding, scaling, and temporal splitting. All preprocessing decisions are grounded in the statistical properties of the data and the domain requirements of procurement integrity analysis.

1 GitHub Repository Architecture

The project repository is publicly accessible[1] and follows the prescribed directory structure exactly.

```
kenya-procurement-triage/
|-- .gitignore
|-- requirements.txt
|-- README.md
|-- data/
|   |-- raw/
|   `-- processed/
|-- phase1_strategy/
|   `-- Phase1_Strategy_Report.pdf
|-- phase2_implementation/
|   `-- notebooks/
|       `-- pipeline_execution.ipynb
|-- src/
|   |-- __init__.py
|   |-- transformers.py
|   `-- pipeline_builder.py
`-- figures/
```

The `.gitignore` excludes raw data, binaries, and virtual environments. All dependencies are pinned in `requirements.txt`.

2 Dataset Overview and Problem Audit

2.1 Data Source

The dataset is downloaded from the Open Contracting Partnership Data Registry, Kenya PPRA[2] publication 147 [1]. It is distributed as a single compressed archive (`kenya_ppra_full.csv.tar.gz`, approximately 55 MB) containing five linked CSV files published to the Open Contracting Data Standard (OCDS)[3] [2]. The data is published as open government data under Executive Order No. 2 of 2018.

2.2 File Structure and Linking Keys

The five files and their row counts are summarized in Table 1. All files link back to `main.csv` via the `main_ocid` column. Awards and suppliers are further linked via `awards_id`.

File	Rows	Columns
main.csv	259,709	15
awards.csv	109,163	9
awards_suppliers.csv	79,870	5
contracts.csv	109,163	12
parties.csv	813,078	16
Merged working table	**277,325**	**20**

Table 1: File sizes and the merged working table ($N = 277{,}325$, $P = 20$).

[1] https://github.com/Razan-O-Elobeid/kenya-procurement-triage

[2] Public Procurement Regulatory Authority (PPRA): the Kenyan government body responsible for overseeing public procurement.

[3] Open Contracting Data Standard (OCDS): an international open-data specification for publishing procurement records: https://standard.open-contracting.org.

The merged table is larger than `main.csv` because some tenders have multiple awards. This is expected.

2.3 Machine Learning (ML) Task Type

This is an **unsupervised risk-ranking** task. No dataset anywhere labels a Kenyan contract as corrupt. Formulating this as supervised classification would require inventing labels, making any reported accuracy meaningless. The target variable Y is therefore a composite risk score engineered from five computable red-flag indicators following the Corruption Risk Index (CRI) methodology [3, 4], each normalized to $[0, 1]$:

$$R(c) = \sum_{k=1}^{K} w_k f_k(c), \quad \sum_{k=1}^{K} w_k = 1 \qquad (1)$$

Equal weights are the default. The five indicators are defined in Section 4.

2.4 Variable Inventory

Table 2 categorizes every column in the merged working table by data type.

2.5 Baseline Missingness Metrics

Table 3 records the missing value profile prior to any processing.

The overall missing data rate across the merged matrix is approximately 31%. Complete rows represent approximately 28% of the dataset. Note on `status`: the 100 recorded terminations almost certainly undercount reality, as contract failures are often not formally documented. The binary flag `is_terminated` remains a valid but conservative risk signal.

Target variable distribution. `value_amount` has mean 15,409,360 KES and median 498,150 KES (ratio 30:1), confirming extreme right skew. The maximum single award is 160.6 billion KES. Five awards record a value of zero, which are data-entry errors. The standard deviation (777 million) exceeds the mean, indicating the distribution spans several orders of magnitude.

3 Cleaning and Imputation Strategy

3.1 Missingness Mechanism Diagnosis

MNAR columns (value missing because of what it would have been). These columns are missing because of what the missing value would have been. Imputing would destroy the signal.

`tender_awardCriteria` (79.3% missing): contracts that skip documenting how a winner was chosen are more likely to be irregular. The missingness is the signal.

`supplier_name` (71.2% missing): awards without a named supplier mean the winning party was not recorded. This is itself a red flag.

`description` in awards and contracts (95.2% missing): absence of a description correlates with incomplete disclosure.

`status` in contracts (99.9% missing): only 100 rows have a value, but "terminated" contracts warrant auditor attention.

Treatment for all MNAR columns: convert each to a binary missing-indicator flag rather than imputing:

- `award_criteria_missing` = 1 if `tender_awardCriteria` is blank, else 0
- `supplier_missing` = 1 if no supplier name recorded, else 0
- `description_missing` = 1 if description is blank, else 0
- `is_terminated` = 1 if status equals "terminated", else 0

MAR columns (missingness linked to another observed column). `contractPeriod_startDate` (34.6% missing) and `dateSigned` (13.6% missing): missing dates are more frequent for direct awards, where formal timelines are bypassed. Missingness is linked to `tender_procurementMethod`, not to the date value itself.

Treatment: median imputation grouped by `tender_procurementMethod`. Imputing with a global median would be incorrect because direct awards have shorter timelines than open tenders; group-based imputation preserves this relationship.

MCAR columns (Missing Completely at Random). `tender_mainProcurementCategory` (0.26% missing): mode imputation. Tender period dates (0.09% missing): median imputation. Both are safe because the gaps are random, and the alternatives (KNN, multiple imputation) add

Column	Type	Role / Notes
ocid, id, tender_id, main_ocid	Nominal Categorical	Join keys only, drop before modeling
date	Datetime	Upload timestamp, used as train/test split key
tag	Nominal Categorical	Always "compiled", drop
language	Nominal Categorical	Always "en", drop
initiationType	Nominal Categorical	Always "tender", drop
buyer_id	Discrete	Buyer group key for concentration features
buyer_name	Nominal Categorical	Buyer identity in plain text
tender_title	Text	Free-text description of what is being purchased
tender_mainProcurementCategory	Nominal Categorical	Three values: goods, works, services
tender_procurementMethod	Ordinal Categorical	open / selective / direct, key red-flag feature
tender_tenderPeriod_startDate	Datetime	When bidding opened
tender_tenderPeriod_endDate	Datetime	Bidding deadline
tender_awardCriteria	Nominal Categorical	79.3% missing, converted to binary flag
value_amount	Continuous	Contract value in KES[4]; 0% missing in awards.csv
value_currency	Nominal Categorical	Always "KES", drop
supplier_name	Text	Supplier identity; 26.8% missing; primary target of entity resolution
contractPeriod_startDate	Datetime	Contract start; 34.6% missing
contractPeriod_endDate	Datetime	Contract end; 10.5% missing
dateSigned	Datetime	Date contract was signed; 13.6% missing
status	Nominal Categorical	99.9% missing, converted to is_terminated flag

Table 2: Variable inventory for the merged working table. "Drop" indicates a constant or identifier column removed before modelling. *supplier_name* is classified as Text because its primary use is string-matching-based entity resolution, not categorical grouping.

Column	File	Missing %	Diagnosed Mechanism
status	contracts	99.9%	MNAR: only "terminated" cases recorded; true termination count is likely higher as governments do not always update records on contract failure
description	awards/contracts	95.2%	MNAR: missing disclosure correlates with irregularity
tender_awardCriteria	main	79.3%	MNAR: contracts hiding selection criteria
supplier_name	awards_suppliers	71.2%	MNAR: awards without a named winner
value_amount (merged)	merged	60.6%	MNAR: tenders with no recorded award
contractPeriod_startDate	awards	34.6%	MAR: linked to tender_procurementMethod
dateSigned	contracts	13.6%	MAR: linked to procurement method
contractPeriod_endDate	awards	10.5%	MAR: linked to procurement method
tender_mainProcurementCategory	main	0.26%	MCAR: small random gaps
tender_tenderPeriod dates	main	0.09%	MCAR: small random gaps

Table 3: Baseline missingness profile and diagnosed mechanisms.

computational cost for negligible gain at these low missingness rates.

3.2 Outlier Identification and Treatment

`value_amount` is right-skewed (mean/median ratio 30:1). The IQR rule is appropriate for skewed distributions:

$$\text{Lower} = Q_1 - 1.5 \times IQR$$
$$\text{Upper} = Q_3 + 1.5 \times IQR$$

Treatment: Winsorization (capping at the 99th percentile), not row deletion. Justification: a row containing an extreme `value_amount` still carries a valid signal in all other columns (procurement method, buyer identity, supplier name). Deleting the row loses that information. Winsorization preserves the row while softening numerical distortion for the model. The Z-score rule was rejected because it assumes a Gaussian distribution; `value_amount` is heavily right-skewed, making Z-scores unreliable for outlier detection here.

Zero-value awards (5 records) are data-entry errors: set to `NaN` and imputed using the median per `tender_mainProcurementCategory`.

Negative tender durations (start date after end date) are data-entry errors from bulk-uploaded historical records: set to `NaN` and imputed using the median per procurement method.

3.3 Entity Resolution Strategy

Supplier names are inconsistent ("GEMINIA INSURANCE COMPANY LIMITED" vs "Geminia Insurance Ltd"). Two-stage resolution:

1. **Primary:** match on `identifier_id` from `parties.csv` where available (68% of rows). Same registration number confirms identical entity.
2. **Fallback:** fuzzy string matching using token sort ratio (threshold ≥ 90) where no identifier exists.

Error rate will be estimated on a manually labeled sample of 200 supplier name pairs (precision and recall reported). Three external sources will validate high-risk supplier matches: the World Bank Debarred Firms list[5], the Kenya Gazette company register[6], and the OpenSanctions PEP[7] database.

4 Feature Engineering

Five risk indicators form the composite score $R(c)$ in Equation 1 [4, 5, 6].

f_1: **is_direct_award.** Binary. 1 if `tender_procurementMethod` is "direct" or "SpeciallyPermitted", else 0. Direct awards bypass competition. 8,744 contracts (3.4%) use this method.

f_2: **award_criteria_missing.** Binary. 1 if `tender_awardCriteria` is blank, else 0. 79.3% of tenders omit this field.

f_3: **tender_duration_days.** Continuous, normalized. Days between tender period start and end. Short duration (< 7 days) gives other suppliers no time to respond.

$$f_3(c) = \texttt{endDate} - \texttt{startDate} \quad \text{(in days)} \tag{2}$$

Negative values (start after end) are treated as data-entry errors, set to `NaN`, and imputed with the median per procurement method before normalization.

f_4: **supplier_concentration.** Continuous, normalized. For each buyer-supplier pair, the fraction of the buyer's total awarded spend that went to this supplier. Values near 1.0 indicate monopoly dependency.

$$f_4(b, s) = \frac{\sum_{c \in C_{b,s}} \texttt{value_amount}(c)}{\sum_{c \in C_b} \texttt{value_amount}(c)} \tag{3}$$

Where $C_{b,s}$ is the set of contracts between buyer b and supplier s, and C_b are all contracts for buyer b.

f_5: **benford_deviation.** Continuous, normalized. The chi-squared distance between the observed first-digit distribution of `value_amount` for a buyer and the theoretical Benford distribution [5].

$$f_5(b) = \sum_{d=1}^{9} \frac{(O_d - E_d)^2}{E_d} \tag{4}$$

Where O_d is the observed proportion of contracts for the buyer b with the first digit d, and $E_d = \log_{10}(1 + 1/d)$ is the Benford expected proportion. High deviation suggests artificially constructed values.

5 Encoding, Scaling, and Transformation Strategy

5.1 Categorical Encoding

Ordinal encoding: `tender_procurementMethod`. Open = 0, selective = 1, direct = 2, SpeciallyPermitted = 3. The order represents increasing restriction on competition. Tree-based models and the isolation forest will recognise this rank. One-hot encoding was rejected because it would discard the meaningful ordering between competition levels.

One-hot encoding: `tender_mainProcurementCategory`. Goods, works, services have no natural order. One-hot encoding avoids imposing a false rank. Three categories produce two binary columns (one dropped to avoid multicollinearity). Ordinal encoding was rejected because no ranking exists among the three categories.

Binary flags for all MNAR columns, as described in Section 3.1.

[5] https://www.worldbank.org/en/projects-operations/procurement/debarred-firms

[6] https://new.kenyalaw.org/gazettes/

[7] Politically Exposed Persons (PEP): individuals who hold or have held prominent public positions. OpenSanctions database: https://www.opensanctions.org/datasets/default/

Constant columns to drop: `language`, `initiationType`, `tag`, `value_currency` (always KES), and all identifier columns used only for joining.

5.2 Numerical Scaling

`value_amount`: log transformation then StandardScaler. The raw distribution spans from zero to 160.6 billion KES, more than eleven orders of magnitude. Log transformation compresses this range. StandardScaler ($\mu = 0, \sigma = 1$) then centers and scales the log-transformed values. The isolation forest uses hyperplane splits; unscaled monetary values would dominate all other features. MinMaxScaler was rejected because it is sensitive to extreme outliers; even after winsorization, large values remain and would compress all other values to near zero under MinMaxScaler.

$$\texttt{value_amount_log} = \log(1 + \texttt{value_amount}) \quad (5)$$

The log1p transformation ($\log(1 + x)$) is used rather than $\log(x)$ to handle zero values safely, since $\log(0)$ is undefined.

Algorithm 1 shows the complete preprocessing pipeline architecture.

Engineered indicators f_1 to f_5 are already normalized to $[0, 1]$ by construction. No further scaling is needed.

Temporal features (`tender_duration_days`, `days_to_signing`) are right-skewed. Apply log1p then StandardScaler, matching the treatment of `value_amount`.

5.3 Feature-to-Target Statistical Test Matching Matrix

Table 4 maps each feature to the correct statistical metric for measuring its association with the composite risk score $R(c)$.
Each row states the chosen metric, the rejected alternative, and the specific reason grounded in the feature's data type and distribution.

5.4 Multicollinearity Protocol

Pairwise Pearson $|r| \geq 0.85$ threshold applied to continuous features. Where two features are collinear, the one with lower feature-to-target mutual information is dropped. The Variance Inflation Factor VIF > 10 used as a secondary check.

A known collinearity candidate is `tender_duration_days` and `days_to_signing`: both are derived from overlapping date fields. This will be verified empirically in Phase 2 using the upper-triangle correlation heatmap and collinear feature scatterplots.

6 Split Strategy and Leakage Prevention

6.1 Temporal Split

The dataset spans July 2018 to August 2026. A random split would allow a model trained on 2024 contracts to see patterns from 2025, which is look-ahead bias.

The dataset is split by **tender end date** rather than upload date to reflect actual procurement event timing. Upload dates were rejected because a bulk upload of historical records in 2024 distorted the upload-date distribution, placing old contracts in the test set. Records from 2018 to 2023 form the training set (158,014 rows, 60.9%) and records from 2024 to 2026 form the test set (101,320 rows, 39.1%). The 375 records outside this window (0.14% of the dataset) are excluded as data-entry anomalies. This reflects the realistic deployment scenario: a model trained on historical data scores new contracts as they arrive.

`TimeSeriesSplit` with 5 folds is used inside the training period for hyperparameter search. Each fold's test set strictly follows its training set in time.

6.2 Group-Aware Separation

Multiple rows can share the same `buyer_id`. Supplier concentration is computed per buyer, so a buyer appearing in both train and test would leak concentration features across the split.

Mitigation: Concentration statistics are computed only on training buyers. For test buyers, concentration is computed independently using only their test-period contracts.

6.3 Transformer Fit Architecture

All preprocessing transformers (imputers, scalers, encoders) are fit strictly on `X_train`, as shown in Algorithm 1. The `sklearn.pipeline.Pipeline` and `ColumnTransformer` wrappers enforce this by design. No transformer sees `X_test` until the evaluation step.

Algorithm 1: Leakage-free preprocessing and scoring pipeline using Isolation Forest [7]. Implemented via `sklearn.pipeline.Pipeline` and `ColumnTransformer`, which enforce that no transformer is refit on X_test.

Input: X_train, X_test
Output: Transformed X_train, scored X_test

Fit on X_train only:
Impute MAR columns (group median by procurement method)
Impute MCAR columns (mode/median)
Winsorise value_amount at 99th percentile
Apply log1p to value_amount, duration, days_to_signing
Fit StandardScaler on log-transformed numerics
Fit OrdinalEncoder on tender_procurementMethod
Fit OneHotEncoder on tender_mainProcurementCategory
Compute supplier concentration from training buyers only
Compute Benford deviation from training values only

Transform X_test (no refitting):
Apply all fitted transformers to X_test
Score with IsolationForest.decision_function(X_test)

6.4 Target Leakage Prevention

The composite risk score $R(c)$ is computed from features derived only from the contract's own record and from training-set aggregates. No future information is used as input. Supplier concentration uses only training-period award history. Benford deviation uses only training-period value distributions.

Table 5 summarizes all planned decisions for the Phase 2 decision log.

Feature	Feature Type	Target Type	Chosen / Rejected / Reason
value_amount_log	Continuous	Continuous	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Even after log transformation, value_amount retains right skew (mean/median ratio 30:1). Pearson r assumes Gaussian; skew violates this and produces misleading correlations. Spearman converts to ranks, making it distribution-free.
tender_duration_days	Continuous	Continuous	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Durations are heavily right-skewed; most tenders last weeks but some last years. Rank-based correlation is appropriate.
supplier_concentration	Continuous	Continuous	**Chosen:** Pearson r. **Rejected:** Spearman ρ. Normalised to $[0, 1]$ by construction and approximately symmetric. Pearson is more precise when the Gaussian assumption is reasonably met.
benford_deviation	Continuous	Continuous	**Chosen:** Pearson r. **Rejected:** Spearman ρ. Same reasoning as supplier_concentration: normalized and approximately symmetric.
days_to_signing	Continuous	Continuous	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Signing delays are right-skewed; most contracts signed quickly but some take unusually long.
is_direct_award	Binary	Continuous	**Chosen:** Point-Biserial r. **Rejected:** Spearman ρ. Ranks of 0 and 1 carry no meaningful information. Point-Biserial r is the mathematically correct form of Pearson r for binary vs continuous variables.
award_criteria_missing	Binary	Continuous	**Chosen:** Point-Biserial r. **Rejected:** Spearman ρ. Same reasoning as is_direct_award.
is_terminated	Binary	Continuous	**Chosen:** Point-Biserial r. **Rejected:** Spearman ρ. Same reasoning as is_direct_award.
tender_procurementMethod	Ordinal Categorical	Continuous	**Chosen:** Spearman ρ. **Rejected:** Pearson r. Encoded as 0–3 but intervals are not equal; Spearman treats values as ranks, matching the ordinal nature of the column.
tender_mainProcurement Category	Nominal Categorical	Continuous	**Chosen:** Mutual Information. **Rejected:** Pearson r, Spearman ρ, ANOVA F. All correlation-based tests require numeric input. Mutual Information works on any data type and captures non-linear dependencies that linear correlation tests miss.

Table 4: Feature-to-target statistical test matching matrix.

Preprocessing Decision	Justification	Method
MNAR missingness	Missingness is itself a risk signal	Binary flag
MAR missingness	Linked to procurement method	Group median imputation
MCAR missingness	Random small gaps	Mode or median imputation
Zero award values	Data-entry errors	Replace with NaN, then impute
Extreme value outliers	Right-skewed; preserve row information	Winsorise at 99th percentile
value_amount transformation	Spans more than eleven orders of magnitude	log1p then StandardScaler
Temporal feature transformation	Right-skewed	log1p then StandardScaler
Engineered indicators	Already in $[0, 1]$ by construction	No scaling needed
tender_procurementMethod	Ordered risk levels	Ordinal encoding 0-3
tender_mainProcurementCategory	No natural order	One-hot encoding
Constant columns	Zero variance, no predictive value	Drop
Train/test split	Tender end date; prevent look-ahead bias	TimeSeriesSplit 60.9/39.1
Buyer concentration leakage	Buyers span both sets	Compute aggregates per set
Transformer fit	Must not see test data	Fit on X_train only via Pipeline

Table 5: Pre-implementation decision log for Phase 2 strategy-vs-reality comparison.

References

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[7] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413–422.